Understanding Network Protocols: The Backbone of Seamless Internet Communication

OSI Model (Open Systems Interconnection Model)

- The OSI Model, developed by the International Standards Organization (ISO), was conceptualized in 1976 and became practically recognized in 1986–1987.
- It serves as a reference model, offering a framework for understanding how data communication occurs in networks. It's important to note that the OSI model itself is not directly implemented in real-world scenarios; instead, the TCP/IP model is widely used in practice.
- The OSI model comprises 7 layers, each with distinct functions, categorized into two main directions of data flow:

1. Down the stack (Sender Side): This process is called Encapsulation, where data is packed with necessary headers for transmission.
2. Up the stack (Receiver Side): This process is called Decapsulation, where headers are removed to interpret the original data.

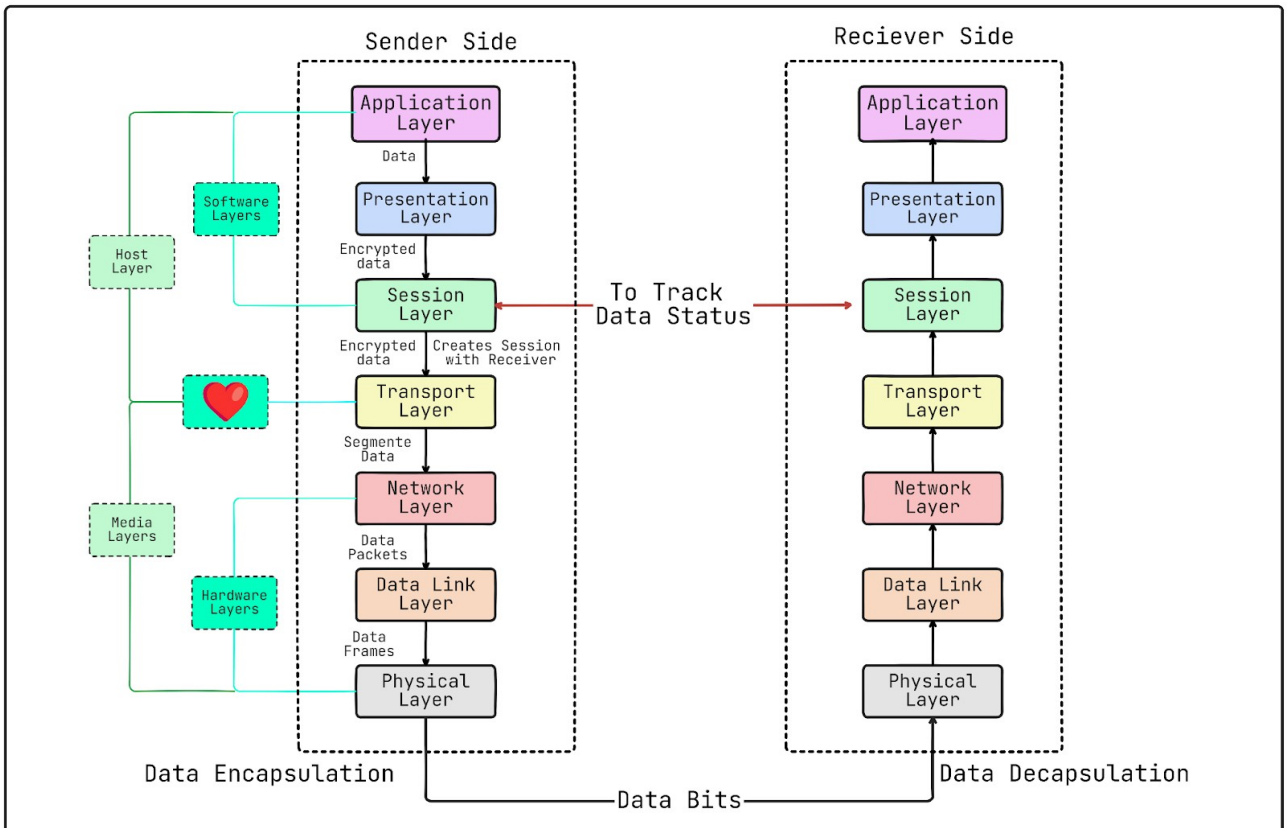

Fig. Layers In OSI Model

1. Application Layer

- This layer provides an interface for users and applications to interact with the network.
- Examples: Email services, web browsers, HTTP, FTP, and DNS.
- Purpose: Enables users to request services and receive responses.

2. Presentation Layer

- Responsible for data translation, encryption, and formatting.
- Converts data into a format understandable by the application layer.
- Examples: Handling file formats such as .mp4, .docx.

3. Session Layer

- Manages the establishment, maintenance, and termination of communication sessions between devices.
- Ensures that data transfer is successful and handles logical ports.
- Once the data is transferred to the receivers, session layer the created session is then ended by the sender side
- Example Functions: Establishing a connection, verifying successful data transmission, and recovering from interruptions.

4. Transport Layer

- Manages data segmentation and ensures reliable delivery of data between devices.
- Works with protocols like TCP (Transmission Control Protocol) and UDP (User Datagram Protocol).
- Responsible for error correction, ensuring all data packets are received in order.
- Note: In the diagram there is a Heart beside this layer indicating this is very important or we can say this layer is the heart of OSI model.

5. Network Layer

- Responsible for routing data packets across networks.
- Determines the best path for data to travel between devices.

6. Data Link Layer

- Handles error detection, flow control, and framing of data packets.
- Ensures that data is properly formatted and ready for transmission via the physical layer.

7. Physical Layer

- Concerned with the actual transmission of raw binary data (0s and 1s) over a physical medium.
- Examples: Copper wires, fiber optics, wireless channels.

Additional Key Concepts

1. Protocol Data Unit (PDU)
   - Each layer communicates using a unit of data called a PDU (Protocol Data Unit).
   - For example, at the transport layer, the PDU can be in the form of a segment or datagram, while at the network layer, it is referred to as a packet.

2. Sender vs. Receiver Perspective
   - Sender Side: The first layer is the Application Layer, where the data originates.
   - Receiver Side: The first layer is the Physical Layer, as this is where the data enters the receiving system.

3. Technical Highlights
   - The OSI model helps conceptualize the communication process but is not directly implemented.
   - Its practical counterpart is the TCP/IP model, which condenses some of the OSI layers for real-world use.

Simplified Flow of Data

1. Sender Side (Encapsulation):
   - Data starts at the Application Layer and travels downward through the layers, with headers added at each step.
   - The physical medium (Physical Layer) transmits the final data as binary (e.g., 01010101).

2. Receiver Side (Decapsulation):
   - Data travels upward through the layers, with headers stripped off at each step until it reaches the Application Layer for interpretation.

TCP/IP Model

- TCP: Transmission Control Protocol (works at the Transport Layer).
- IP: Internet Protocol (works at the Network Layer).
- Facilitates communication between devices over a network.

Key Features of TCP/IP:

- **End-to-End Control:** Ensures all transmitted data is received by the destination.
- **Decentralism:** No centralized control; endpoints handle the verification of data transfer.
- **Dynamic Routing:** Chooses the best path dynamically for data transmission.

Common Misconceptions:

- TCP/IP ≠ Advancement of OSI:
- They are separate models with no direct dependency.
- Both were developed to address the lack of communication between devices from different manufacturers.

Historical Background:

- Early networking devices (e.g., Dell to Dell, Cacio to Cacio) could only communicate within the same brand. TCP/IP and OSI models were introduced to solve such issues and facilitate global interoperability.

Comparison of OSI and TCP/IP

OSI	TP IP
Developed by ISO (International Organization for Standardization)	Developed by the American Defense Department.
Developed by ISO (International Organization for Standardization)	Practically implemented and used for internet communication.

Layers of the TCP/IP Model

4 - Layer TCP/IP Model Architecture

1. Application Layer:

- Protocols: HTTP, FTP, TFTP, SMTP, DHCP, etc.
- Function: Provides a user interface for communication.

2. Transport Layer:

- TCP: Reliable but slower, suitable for non-real-time communication.
- UDP: Faster but less reliable, used for real-time communication (e.g., voice calls).

3. Network Layer:

- Protocols: ICMP, IP, ARP, RARP.
- Function: Logical data transmission; includes sender/receiver IP addressing.

4. Network Access Layer:

- Combines the Physical and Data Link layers of the OSI model.
- Manages hardware and MAC addressing.

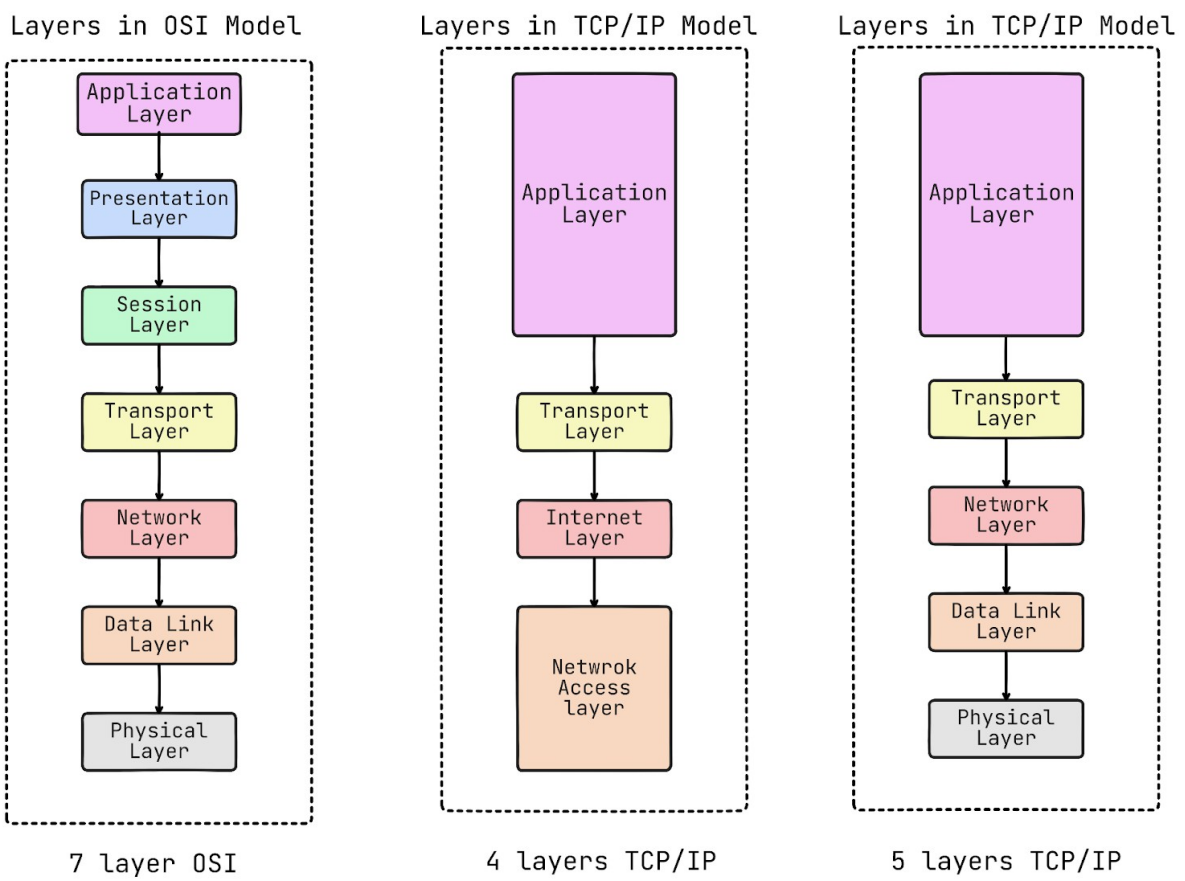

Fig. Direct View of OSI and TCP/IP Models

Issues Solved by TCP/IP Model

- Addressing: Manages IP and MAC addresses effectively.
- Routing: Determines data paths dynamically.
- Domain Name Resolution: Matches domain names to IP addresses.
- Flow and Error Control: Ensures smooth and error-free communication.
- Interoperability: Allows devices from different manufacturers to communicate.

Protocols Used In TCP/IP Model

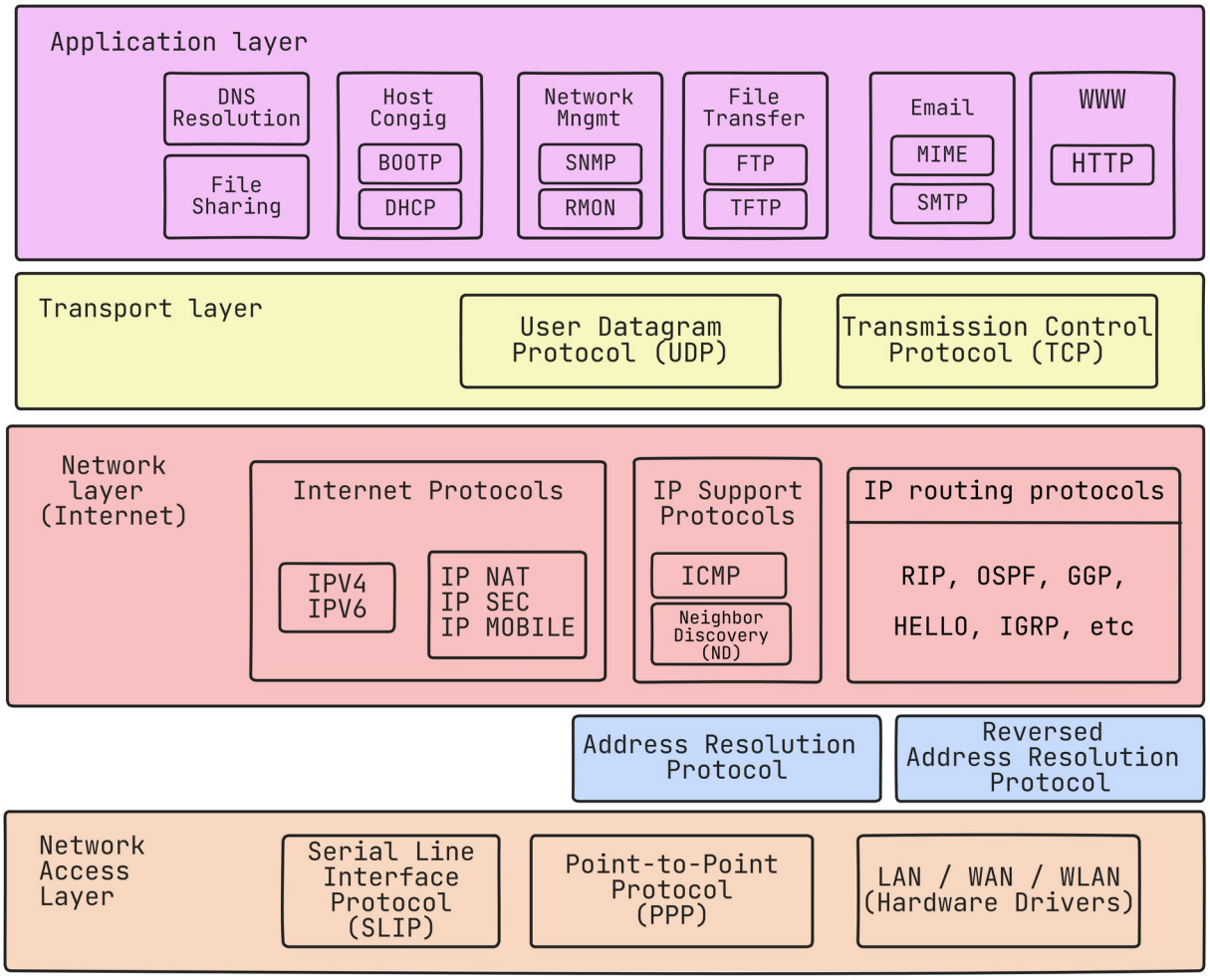

Fig. Protocols used in TCP/IP Models